MIL-T-37131
27 March 1975
SEE 6.3 FOR
SUPERSESSION DATA

MILITARY SPECIFICATION

# TROUSERS, OPERATING, SURGICAL, MAN'S, COTTON, GREEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for one type and four sizes of a green, cotton, man's, surgical, operating trousers.

1.2 Classification. The trousers shall be furnished in the following sizes, as specified (see 6.2):

SIZES

| Small | Medium | Large | Extra Large |
|---|---|---|---|

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| V-T-276 | Thread, Cotton |
| UU-P-268 | Paper, Kraft, Wrapping |
| CCC-C-444 | Cloth, Jean, Cotton |
| DDD-L-20 | Label, For Clothing, Equipage, and Tentage, (General Use) |
| PPP-B-636 | Boxes, Shipping, Fiberboard |

FSC 6532

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STANDARDS

Federal

| | |
|---|---|
| FED-TEST METHOD STD. NO. 191 | Textile Test Methods |
| FED-STD-751 | Stitches, Seams, and Stitchings |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Color Association of the United States, Incorporated

TCA Cable Numbers and Colors

(Application for TCA Cable numbers and colors should be addressed to the Color Association of the United States, Incorporated, 200 Madison Avenue, New York, New York 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials and components. The materials and components shall conform to the applicable specifications and standards as listed or required herein.

3.1.1 Fabric.

3.1.1.1 Trousers. The trousers shall be fabricated from the cotton jean cloth conforming to CCC-C-444, Class 2, dyed to approximately match Grayish Green 180, except for the following:

a. The pH value of the water extract of the finished fabric shall be 5.0 to 9.0.

b. The dyed and finished fabric shall show at least "fair" colorfastness to light when exposed for 40 standard hours.



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3.1.1.2 Waistband lining and reinforcement. The waistband lining and reinforcement shall be fabricated from the trousers cloth, 3.1.1.1.

3.1.2 Thread. All the sewing operations shall be performed with the cotton thread conforming to V-T-276, Type 1A3 or 1B3, Ticket Number 50, 3 ply and Ticket Number 70, 2 ply (see Table II). When Stitch Type 401, is utilized, Ticket Number 50, 3 ply may be used for the bobbin or looper thread instead of Ticket Number 70, 2 ply. The color of the thread shall approximately match the color of the trousers fabric, 3.1.1.1. The thread shall show "good" colorfastness to light, laundering and perspiration.

3.1.3 Drawcord. The drawcord shall be fabricated from tape knitted on a Raschel warp knitting machine. The sides of the tape shall be non-ravelling. The knitted tape shall conform to Table I when inspected as specified in 4.3.1.

TABLE I
DRAWCORD KNITTED TAPE PROPERTIES

| CHARACTERISTICS | REQUIREMENTS |
|---|---|
| Material | Cotton |
| Total Wales, minimum | 8 |
| Courses per inch, minimum | 8 |
| Width, Inches | 5/8 ± 1/16 |
| Breaking strength, pounds minimum | 65 |
| Ounces per gross yards, minimum | 20 |
| Color | Bleached white |
| Length, inches | 60 ± 1/2 |
| Finish of each end | Self fringes |
| Length of fringes, inches | One to 1-1/2 |

3.2 Design. The trousers shall be of the pajama design, without fly and pockets, crossed drawcord waist and hemmed leg bottoms. The trousers shall conform in appearance to Figure 1.

3.3 Construction.

3.3.1 Cutting. The trousers shall be cut in strict accordance with the patterns furnished by the Government (see 6.4), which provide for seam allowance, directional lines for cutting and notches for the proper assembling of all the parts. The Government patterns shall not be altered in any manner and shall be used for making the working patterns. The working patterns shall be identical to the Government furnished patterns. The component parts shall be cut from the same ply of the lay. The waistband lining shall be cut in the direction of the warp. The waistband lining shall be of sufficient width and length to conform to the requirements of operation 6. The lining may be strip cut to proper width and placed on rolls or may be cut from the ends. The waistband lining may be one continuous piece or two pieces.



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3.3.2 Shade and size marking. Except for the waistband lining which may be cut from the ends, the component parts of the trousers shall be marked or ticketed by any commercial method to insure a uniform shade and proper assembly throughout the trousers. No metal fastening device or sewn-on ticket shall be used. The shade and size markings shall be accomplished by the use of an ink pad numbering machine, rubber stamp, or pencil, provided the markings do not show through to the outerside of the trousers and are not deleterious to the cloth. The markings, which are placed on the seam allowance, shall not be visible on the outerside. The adhesive type shade and size marking tickets may be used for the shade and size markings, provided they conform to 3.3.2.1. The assembled trousers shall not contain any shade or size marking ticket nor any marking that was used for locating the position for attaching of any component.

3.3.2.1 Adhesive shade and size marking tickets. The adhesive shade and size marking tickets shall be fabricated from paper and shall contain a thermo-activated type adhesive on one side. The adhesive shall not discolor or damage the cloth and the adhesive mass shall not adhere to the cloth, upon removal of the ticket. The heat used to attach the thermoactivated adhesive tickets shall not stiffen, scorch, or damage the cloth in any manner.

3.3.3 Replacement of defective components and cut parts. Care shall be exercised during the spreading, cutting and manufacturing processes to assure that any component and cut part containing any material defect or damage that is classified in this specification as a defect shall be replaced with a non-defective and properly matched component or cut part, as applicable.

3.3.4 Seams and stitching. All the seams and stitching used in the fabrication of the trousers shall conform to Table II. The seam and stitch types shall conform to FED-STD-751. When two or more seam or stitch types are specified for the same part of the operation, the contractor may select one of them for the operation. The minimum and maximum number of stitches for each operation shall conform to Table II. When Stitch Type 401 is used, the looper thread shall be on the innerside of the garment. The gage of the stitching for all the double lapped, double-stitched seams shall be 3/16 to 1/4 inch. The outer row of the stitching shall be 1/16 to 1/8 inch from the seam edge. The seam allowances shall be maintained so that no runoff or open seam shall occur. All seams shall start and finish uniformly. No seam or component shall be twisted, puckered, or pleated and no part of the trousers shall be caught in an unrelated operation or seam. All the seam edges shall be properly forced out and shall not contain any folds. The waistband lining shall be uniform in width, smooth and flat. All the thread ends shall be trimmed to a maximum length of 1/4 inch. Except for the ends of the drawcord, no cut or raw edge shall be visible on the interior or exterior of the trousers.



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3.3.4.1 Sewing. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching, in any row, shall be maintained within a tolerance of plus or minus 1/32 inch. The same tolerance for spacing and parallelism shall be applicable for the double rows of stitches. The ends of Stitch Type 301 stitching that are not caught in other seams or stitching shall be backstitched, by overlapping on themselves, a minimum of 1/2 inch. Each thread break, skip, and run-off shall be overstitched not less than 1/2 inch. Skipped stitches or thread breaks on Stitch Type 401 stitching may be repaired using Stitch Type 301. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together.

3.3.5 Manufacturing operation requirements. The trousers shall be manufactured in accordance with the operation requirements specified in Table II. The supplier is not required to follow the exact sequence of operations listed, unless otherwise specified. Any additional basting or holding stitches used to facilitate the fabrication of the trousers are permissable, provided the thread does not show on the surface of the assembled trousers.



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TABLE II

| NO. | MANUFACTURING OPERATIONS REQUIREMENTS | STITCH TYPE | SEAM & STITCHING TYPE | STITCHES PER INCH | THREAD | |
|---|---|---|---|---|---|---|
| | | | | | NEEDLE | BOBBIN OR LOOPER |
| 1. | Join the fronts | | | | | |
| | a. Join the foreparts at the front with a double-lapped, double-stitched seam. The right front shall be over the left front. | 301 | LSc-2 | 10-14 | 50/3 | 50/3 |
| | | or 401 | LSc-2 | 10-14 | 50/3 | 70/2 |
| 2. | Join the backs | | | | | |
| | a. Join the back parts at the back with a double-lapped, double-stitched seam. The seams may lap in either direction. | 301 | LSc-2 | 10-14 | 50/3 | 50/3 |
| | | or 401 | LSc-2 | 10-14 | 50/3 | 70/2 |
| 3. | Join the outseams and inseams | | | | | |
| | a. Join the forepart to the back part at the outseams and inseams with a double-lapped, double-stitched seam. The seams may lap in either direction. The inseams at the crotch, shall be staggered not more than 3/8 inch. | 301 | LSc-2 | 10-14 | 50/3 | 50/3 |
| | | or 401 | LSc-2 | 10-14 | 50/3 | 70/2 |
| 4. | Hem the bottom of the legs | | | | | |
| | a. Fold under the bottom of the trouser leg, in accordance with notches on pattern, with the raw edges turned under to form a hem 7/8 to 1-1/8 inches wide. The stitching shall be 1/16 to 1/8 inch from the edge. | 301 | EFb-1 | 10-14 | 50/3 | 50/3 |
| 5. | Tunnel entrances left and right forepart | | | | | |
| | a. The tunnel entrances shall be straight vertical buttonholes, barred at each end. The stitching shall be securely caught in the fabric. The opening shall be 7/8 ± 1/8 inch. The edges of the buttonholes shall not be ragged. The buttonholes shall be reinforced with a piece of the trousers fabric. | Button-hole | | 44 to 64 per button-hole including bars | 50/3 | 50/3 |
| | b. The buttonholes shall be located 1-1/2 to 1-3/4 inches from the center of the front and 7/8 to 1/2 inch from the top of the trousers. | | | | | |



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TABLE II. (Continued)

| NO. | MANUFACTURING OPERATIONS REQUIREMENTS | STITCH TYPE | SEAM & STITCHING TYPE | STITCHES PER INCH | THREAD | |
|---|---|---|---|---|---|---|
| | | | | | NEEDLE | BOBBIN OR LOOPER |
| 6. | Attach the waistband lining (forming tunnel) and insert drawcord | | | | | |
| | a. The waistband lining may be one continuous piece or may be fabricated from two pieces joined together with one row of stitches approximately 1/4 inch from the edges. The seam shall be flat and smooth. | 301 | SSa-1 | 10-14 | 50/3 | 50/3 |
| | | or 401 | SSa-1 | 10-14 | 50/3 | 70/2 |
| | b. Place the waistband lining on the top inside of the trousers with the ends overlapping at the back seam of the trousers. Fold under the top edge of the trousers and the upper and lower sides of the waistband lining. The overlapping waistband lining end shall also be folded under. Attach the waistband lining to the top of the trousers with two rows of stitching, 1-1/2 ± 1/8 inches apart. Each row of stitching shall be 1/16 to 1/8 inch from the folded sides. The lining shall not be exposed beyond the top of the trousers. | 301 | LSk-2 | 10-14 | 50/3 | 50/3 |
| | | or 401 | LSk-2 | 10-14 | 50/3 | 70/2 |
| | OR | | | | | |
| | c. Place the waistband lining on the top outerside of the trousers so that the upper side edge of the lining and the top edge of the trousers coincide. The overlapping end of the lining shall be folded under. Stitch together, approximately 1/4 inch from the edges, turn and top stitch to the trousers 1/16 to 1/8 inch from the folded side of the waistband lining. | 301 | SSe-2(a) | 10-14 | 50/3 | 50/3 |
| | | or 401 | SSe-2(a) | 10-14 | 50/3 | 70/2 |
| | | and 301 | SSe-2(b) | 10-14 | 50/3 | 50/3 |
| | d. Fold under the bottom side of the waistband lining and attach to the trousers with the stitching 1/16 to 1/8 inch from the bottom side of the lining | 301 | LSd-1 | 10-14 | 50/3 | 50/3 |



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TABLE II (Continued)

| NO. | MANUFACTURING OPERATIONS REQUIREMENTS | STITCH TYPE | SEAM & STITCHING TYPE | STITCHES PER INCH | THREAD | |
|---|---|---|---|---|---|---|
| | | | | | NEEDLE | BOBBIN OR LOOPER |
| 6. | Continued | | | | | |
| | e. Insert the drawcord into the tunnel formed by the waistband lining with the ends crisscrossed through the buttonholes at the top of the trousers. | | | | | |
| | f. Attach the folded overlapping end of the waistband lining to the trousers with one row of stitching 1/16 to 1/8 inch from the folded end. The stitching shall also catch the drawcord, anchoring it in place. | | | | | |
| 7. | Attach label | | | | | |
| | a. Position the identification label on the back inside of the trousers, a maximum of 1/2 inch from the seat seam and not more than 1/4 inch below the waistband lining. Attach on all four sides. When a heat attachable label is used, it may be attached with heat. | 301 | LSbj-1 | 10-14 | 50/3 | 50/3 |
| | b. The top edge of the label may be caught in the bottom seam of the waistband lining. Attach the remaining three sides. | 301 | LSbj-1 | 10-14 | 50/3 | 50/3 |
| 8. | Trim all thread ends and remove all thread scraps and foreign matter. | | | | | |
| 9. | Fold each pair of trousers to an approximate size of 15-1/2 by 10 inches without creating any wrinkles or creases. | | | | | |



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3.4 Dimensions. The trousers after assembly shall conform to the dimensions for the different sizes specified in Table III when measured as specified in 4.3.6. The variation in length between the left and right leg inseam dimension shall not exceed 1/2 inch in any pair of trousers.

TABLE III
TROUSERS DIMENSIONS IN INCHES

| Location | Size | | | | Tolerance |
|---|---|---|---|---|---|
| | Small | Medium | Large | Extra Large | |
| Waist | 17-7/8 | 19-5/8 | 21-5/8 | 23-3/4 | ± 1/2 |
| Seat | 22 | 23-1/2 | 25 | 26-1/2 | ± 1/2 |
| Inseam | 31-3/4 | 32-3/4 | 33-7/8 | 33-7/8 | ± 1/2 |
| Outseam | 42-3/8 | 43-7/8 | 45-3/8 | 45-7/8 | ± 1/2 |
| Leg Bottom | 10 | 10-1/4 | 10-1/2 | 10-3/4 | ± 1/4 |

3.5 Label. The identification label for each trousers shall conform to DDD-L-20, Type I, Class 1, Launderable Label, except that the color of the label shall be one of the following colors for the appropriate trousers size:

| COLOR | TCA CABLE NUMBER | TROUSERS SIZE |
|---|---|---|
| Peach Blossom | 70098 | Small |
| Lemon Yellow | 70205 | Medium |
| Indian Orange | 70072 | Large |
| Primitive Green | 70167 | Extra Large |

When inspected as specified in 4.3.3, the color of the label shall approximately match the crepe side of the applicable TCA Cable Number. The color of the label shall be obtained with vat or naphthol dyes and shall show at least "good" colorfastness to laundering transference and at least "fair" colorfastness to accelerated laundering. The color of the printing medium shall be black. The label shall be a heat attachable or a sew-on type. When heat attachable labels are used, the back of the label shall contain a thermoplastic compound suitable for being attached to the trousers by heat. When adhered to the trousers, the label shall remain completely adhered without any evidence of separation when inspected as specified in 4.3.2. The size of the label shall be adequate to contain the following information:

TROUSERS, OPERATING, SURGICAL, MAN'S,
COTTON, GREEN
MIL-T-37131
SIZE
CONTRACT OR ORDER NUMBER
NSN



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3.5.1 Alternate. An alternate identification label fabricated from cotton coutil or poplin cloth having a greige weight of 2.11 yards per pound (45 inch width) may be used in lieu of the label specified in 3.5. The printing medium and printing shall conform to DDD-L-20, Type I, Class 1, Launderable Label. The color and colorfastness of the label shall be as specified in 3.5. The label shall contain the same information as specified in 3.5.

3.6 Workmanship. The trousers shall not contain any spot, stain, hole, abraded area, tear, cut, mend patch, scorch, or needle chew. The trousers shall conform to the quality of grade established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material and component inspection. In accordance with 4.1, the materials and components shall be inspected and tested in accordance with all the requirements of the referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification or purchase document.

4.1.2 Certificate of quality. When certificates of quality are submitted in accordance with Table IV, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the components to verify the validity of the certification.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (an investigation without the use of special laboratory apparatus or procedures such as visual) and testing (a determination by technical means of the physical and chemical properties) of the item.



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4.2 Quality conformance inspection. The sampling and inspection shall conform to MIL-STD-105, except where otherwise indicated hereinafter. The quality conformance inspection shall consist of the tests and examinations specified in Table IV.

4.2.1 Sampling.

4.2.1.1 Inspection lot.

4.2.1.1.1 Drawcord tape. An inspection lot size shall be expressed in units of one linear yard of the drawcord tape and shall consist of all the drawcord tape received by the trousers manufacturer at one time. The sample unit for inspection shall be one linear yard.

4.2.1.1.2 Labels. An inspection lot size shall be expressed in units of one label and shall consist of all the labels received by the trousers manufacturer at one time. The sample unit for inspection shall be one label.

4.2.1.1.3 Alternate identification label fabric. An inspection lot size shall be expressed in units of one linear yard of the alternate identification label fabric and shall consist of all the alternate identification label fabric received by the trousers manufacturer at one time. The sample unit for inspection shall be one linear yard of the alternate identification label fabric.

4.2.1.1.4 Trousers. An inspection lot size shall be expressed in units of one pair of trousers made essentially under the same conditions and from the same materials and components. The sample unit for inspection shall be one pair of trousers.

4.2.1.1.5 Preparation for delivery. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing pairs of trousers of one size, fully prepared for delivery from essentially the same materials and components. The sample unit for inspection shall be one shipping container, containing pairs of trousers of one size, fully prepared for delivery with the exception that it need not be sealed.

4.2.1.2 Sampling for tests and examinations of the drawcord tape, labels, alternate identification label fabric, trousers, and preparation for delivery. The sample size, acceptance criteria, tests and examinations required for the drawcord tape, labels, alternate identification label fabric, trousers, or the preparation for delivery, as applicable, shall be as specified in Table IV.



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TABLE IV
SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS OF THE DRAWCORD TAPE, LABELS, ALTERNATE IDENTIFICATION LABEL FABRIC, TROUSERS AND PREPARATION FOR DELIVERY

| Item | Inspection | Paragraph Requirement | Paragraph Method | Sample Size | Acceptance Criteria | |
|---|---|---|---|---|---|---|
| Drawcord Tape 1/ | Material | 3.1.3 | 4.3.1 | 2/ | 2/ | 4/ |
| | Total wales | 3.1.3 | 4.3.1 | 2/ | 2/ | 4/ |
| | Courses per inch | 3.1.3 | 4.3.1 | 2/ | 2/ | 4/ |
| | Width | 3.1.3 | 4.3.1 | 2/ | 2/ | 4/ |
| | Breaking strength | 3.1.3 | 4.3.1 | 2/ | 2/ | 4/ |
| | Ounces per gross yards | 3.1.3 | 4.3.1 | 2/ | 2/ | 4/ |
| Labels | Adherance 5/ | 3.5 | 4.3.2 | 3/ | 3/ | 4/ |
| | Color comparison 6/ | 3.5 | 4.3.3 | 3/ | 3/ | 4/ |
| Alternate Identification Label Fabric 1/ | Cotton identification | 3.5.1 | 4.3.4 | 2/ | 2/ | 4/ |
| | Weight | 3.5.1 | 4.3.4 | 2/ | 2/ | 4/ |
| | Kind of fabric | 3.5.1 | 4.3.4.1 | 2/ | 2/ | 4/ |
| Trousers | Visual examination | 3.2 through 3.3.5 and 3.5 through 3.6 | 4.3.5.1 | Inspection Level II | An acceptable quality level of 2.5 defects per 100 units for Major defects and 10.0 defects per 100 units for total (major plus minor) defects | |
| | Dimensional check | 3.4 | 4.3.5.1 | Inspection Level S-2 | An acceptable quality level of 4.0 defects per 100 units | |
| | Preparation for delivery | Section 5 | 4.3.5.2 | Inspection Level S-2 | An acceptable quality level of 2.5 defects per 100 units | |



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1/ The inspection of the drawcord tape, labels and the alternate identification label fabric, as applicable, shall be conducted either at the contractor's or subcontractor's plant.

2/ The sampling for the testing of the drawcord tape and the alternate identification label fabric shall be in accordance with 4.2.1.1.1 and 4.2.1.1.3, as applicable, and as follows:

| Lot Size (Yards) | Sample Size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

The samples shall be taken from the beginning of each roll. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified in Table IV. Each sample unit taken for inspection shall be used to determine all the characteristics of the drawcord tape and the alternate identification label fabric specified in Table IV.

3/ The sampling for the testing of the labels shall be in accordance with 4.2.1.1.2 and as follows:

| Lot Size (Number of labels) | Sample Size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

The lot shall be unacceptable if one or more of the sample units fail to meet the adherence or color comparison requirement, as applicable.

4/ A certificate of quality shall be submitted and will be acceptable for the stated requirement (see 4.1.2).

5/ This inspection is only applicable to the labels with thermoplastic backing.

6/ This inspection is applicable to all the kinds of labels.



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4.3 Inspection methods.

4.3.1 Drawcord tape. The drawcord characteristics shall be determined in accordance with the inspection methods of FED-TEST METHOD STD. NO. 191 specified in Table V.

TABLE V
DRAWCORD TAPE PROPERTIES

| CHARACTERISTICS | FED-TEST METHOD STD. NO. 191 METHOD |
|---|---|
| Cotton identification | 1200 |
| Total wales | 5070 |
| Courses per inch | 5070 |
| Weight | 5040 |
| Breaking strength | 5102 |

4.3.2 Adherence. The label with thermoplastic back shall be attached to the trousers fabric with heat in accordance with the instruction of the manufacturer of the label with thermoplastic backing. The attached label shall then be laundered for eight consecutive times in accordance with FED-TEST METHOD STD. NO. 191, Method 5605. Upon completion of the eighth laundering, the label shall be examined for any evidence of separation from the fabric (see 3.5).

4.3.3 Color matching. The color of the labels shall be compared to the creped side of the approved shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500 degrees Kelvin.

4.3.4 Alternate identification label fabric. The characteristics of the alternate identification label fabric shall be determined in accordance with the inspection methods of FED-TEST METHOD STD. NO. 191 specified in Table VI and 4.3.5.1.

TABLE VI
ALTERNATE IDENTIFICATION LABEL FABRIC PROPERTIES

| Characteristic | FED-STD-191 Method |
|---|---|
| Cotton Identification | 1200 |
| Weight | 5041 |

4.3.4.1 Kind of alternate identification label fabric. The kind of fabric used for the alternate identification label shall be determined by any standard commercial method.



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4.3.5 Visual examination.

4.3.5.1 Trousers. Each pair of trousers selected as specified in Table IV as a sample unit from the lot shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification and list of defects, Tables VII and VIII, as applicable, shall be used to classify and enumerate the defects found.

TABLE VII
CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION OF THE ASSEMBLED TROUSERS

| DEFECT | MAJOR | MINOR |
|---|---|---|
| GENERAL | | |
| a. Any hole, scissors or knife cut, tear, mend, burn, or weakening defect such as smash, multiple float, loose slub, needle chew, or abraded area | X | |
| b. Any portion of the trousers stiffened or scorched by any process of manufacturing | X | |
| c. Any missing or broken yarn, thin place, misweave, woven in waste, loose yarn, snagged thread, area of no dye penetration, or dye streak on the exterior of the trousers | 1/ | |
| SHADED PARTS | | |
| a. Any part shaded or any shade bar | | X |
| b. Color of any component not as specified or not uniform | | X |
| CLEANNESS | | |
| a. Three or more thread ends not trimmed to a maximum length of 1/4 inch or three or more thread scraps not removed | | X |
| b. Any unsightly slub, spot, or stain; any discoloration of the cloth caused by the removal of any shade and size marking ticket; any shade and size marking ticket, excluding the metal fastening or sewn-on kind, not removed; any shade and size marking identification stamping visible on the outerside of the trousers; or any shade and size marking adhesive mass adhering to the fabric. | | X |
| c. Any metal fastening device or sewn-on shade and size marking ticket attached to the trousers | X | |



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TABLE VI (Cont'd)

| DEFECT | MAJOR | MINOR |
|---|---|---|
| COMPONENTS AND ASSEMBLY | | |
| a. Any component not as specified or any defect of component or defect of assembly, not herein classified | 1/ | |
| b. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified | 1/ | |
| c. Any dimension not as specified, not herein classified | 1/ | |
| SEAMS AND STITCHING | | |
| a. Accuracy of seaming: | | |
| 1. Any seam, hem, or attachment of any component twisted, puckered, or pleated; any part of the trousers or of any component caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold | | X |
| 2. Ends of stitching that are not caught in other seams or stitching: | | |
| (a) Any not securely backstitched | X | |
| (b) Any secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself | | X |
| 3. Thread breaks: | | |
| (a) Any not securely overstitched | X | |
| (b) Any secure but overstitched for a distance less than 1/2 inch | | X |
| 4. Ends of any leg inseam at the crotch staggered by more than 3/8 inch | | X |



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TABLE VII (Cont'd)

| DEFECT | MAJOR | MINOR |
|---|---|---|
| SEAMS AND STITCHING (Cont'd) | | |
| b. Gage of stitching 2/: | | |
| 1. Any stitching irregular, unevenly or not gaged as specified, or not uniform | | X |
| c. Open seams 3/: | | |
| 1. Other than on the hem: | | |
| (a) Any more than 1/8 inch but not more than 1/4 inch | | X |
| (b) Any more than 1/4 inch | X | |
| 2. On the hem: | | |
| (a) Any more than 1/4 inch but not more than 1/2 inch | | X |
| (b) Any more than 1/2 inch | X | |
| d. Raw edges 4/: | | |
| 1. Outerside or innerside of the trousers: | | |
| (a) Any more than 1/4 inch but not more than 1/2 inch | | X |
| (b) Any more than 1/2 inch | X | |
| e. Seam and stitch type: | | |
| 1. Not specified seam or stitch | 1/ | |
| 2. Any looper thread on the outerside of any trousers seam, when Stitch Type 401 is used | X | |
| f. Stitch tension: | | |
| 1. Any loose stitching 5/ | 1/ | |
| 2. Any tight stitching, resulting in breakage of the stitches, when normal pull is applied 6/ | X | |



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TABLE VII (Cont'd)

| DEFECT | MAJOR | MINOR |
|---|---|---|
| SEAMS AND STITCHING (Cont'd) | | |
| g. Stitches per inch 7/: | | |
| 1. Stitches more or less than specified: | | |
| (a) Trousers (excluding label): | | |
| (1) One or 2 | | X |
| (2) More than 2 | X | |
| (b) Label: | | |
| (1) One to 3 | | X |
| (2) More than 3 | X | |
| h. Buttonholes (tunnel entrances): | | |
| 1. Any omitted or uncut | X | |
| 2. Any misplaced, not located as specified, not specified kind, any end not barred or less than 5/8 inch or more than one inch in length | 1/ | |
| 3. Incomplete stitching, stitching not securely caught in fabric causing stitching to pull away, ragged edge, one or more broken stitches or two or more skipped stitches in any buttonhole | X | |
| DETAILED DEFECTS APPLICABLE TO SPECIFIC PARTS OF THE TROUSERS | | |
| WAISTBAND LINING | | |
| a. Short, tight, or twisted causing puckering | X | |
| b. Width of the lining more than 1-3/4 inches or less than 1-1/4 inches | | X |
| c. Lining exposed beyond the top of the trousers 1/8 to 1/4 inch | | X |
| d. Lining exposed beyond the top of the trousers by more than 1/4 inch | X | |



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TABLE VII (Cont'd)

| DEFECT | MAJOR | MINOR |
|---|---|---|
| DRAWCORD | | |
| a. Less than 59-1/2 inches or more than 61 inches | X | |
| b. Width less than 7/16 inch or more than 9/16 inch | | X |
| c. One or two fringes missing from any end | | X |
| d. More than two fringes missing from any end | X | |
| e. Length of fringes on any end less than one inch or more than 1-1/2 inches | | X |
| f. Ends of drawcord not crisscrossed through the buttonholes | | X |
| HEM AT THE BOTTOM OF THE LEGS | | |
| a. Width of any less than 7/8 inch or more than 1-1/8 inches | | X |
| LABELS | | |
| a. Identification: | | |
| 1. Size missing, incorrect, or illegible or color of any label not as specified | X | |
| 2. Other than the size information, incomplete, incorrect, or illegible | | X |
| 3. Misplaced, off center of the back by more than 1/2 inch | | X |
| 4. Stitching omitted on one or more edges | | X |
| 5. When applicable, any evidence of separation of any heat attached label | | X |



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1/ The defect shall be classified and scored as a major defect, when it seriously affects the serviceability or appearance, otherwise it shall be classified and scored as a minor defect.

2/ The gage of stitching defect shall be scored only when the condition exists for more than 2 inches on any row of stitching.

3/ A seam shall be classified and scored as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be classified and scored as open when either one or both sides of the seam contain any of the aforementioned for a single stitched seam.

4/ Any edge that should be turned under and is not, but is securely caught in the stitching, shall be classified and scored as raw edge. Any raw edge, that is not securely caught in the stitching, shall be classified and scored as an open seam.

5/ A seam shall be classified and scored as loose when the top and bobbin threads are on the same surface.

6/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching, by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified and scored as an accuracy of seaming defect.

7/ The stitches per inch defect shall be scored only when the condition exists for more than 3 inches on any row of stitching.

TABLE VIII
LIST OF DEFECTS FOR THE DIMENSIONS OF THE ASSEMBLED TROUSERS

| ASSEMBLED MEASUREMENT | DEFECT |
|---|---|
| Measure the waist, seat, inseam, outseam and leg bottom. | Any measurement deviating from the measurements specified in 3.4 and Table III and applicable tolerance shall be enumerated as a dimensional defect. |
| Equal Length<br>Leg inseams | The variation in length between the left and right leg inseam shall be enumerated as a dimensional defect when it exceeds 1/2 inch. |



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4.3.5.2 Preparation for delivery. Each of the fully prepared shipping containers, containing trousers of one size, selected as specified in Table IV as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table IX, shall be used to enumerate the defects.

TABLE IX

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

| ITEM | DEFECT |
|---|---|
| Exterior and interior markings | Missing, incorrect, incomplete, or illegible; improper size, location, sequence, or method of application not as specified. |
| Materials | Any non-conforming component; any component or component part missing, damage, or otherwise defective. |
| Workmanship | Inadequate application of the components such as incomplete closure of any exterior container, container flap, or loose strapping; bulging or distortion of any container; any polyethylene bag damaged, any vent hole missing, or any open or non-continuous heat sealed seam. |
| Exterior and interior weight or content | Gross or net weight exceeds the requirement; number per container more or less than specified; more than one size pair of trousers in the same container; any portion of the trousers caught in any heat sealed seam; any pair of trousers distorted, deformed, or bent; any not individually packaged within the polyethylene bag. |

4.3.6 Procedure for measuring. The trousers to be measured shall be laid flat, without tension, on a smooth, flat surface so that any wrinkles or creases shall not affect the measurement. The measurements shall be made with the use of a yard stick, metal tape, or by use of any suitable measuring device. The measurements shall be made as follows:



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a. The waist dimension shall be taken along the top edge of the waistband from side folded edge to side folded edge.

b. The seat dimension shall be taken at a point 3 inches above the crotch seam from side folded edge to side folded edge.

c. The inseam dimensions shall be taken from the crotch seam to the bottom of the leg along the inseam.

d. The outseam dimension shall be taken from the top of the trousers to the bottom of the leg along the outseam.

e. The leg bottom dimension shall be taken at the bottom of the leg from side folded edge to side folded edge.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. Each pair of trousers of the size specified constitutes one unit of issue.

5.1.2 Level A. Each pair of trousers shall be neatly folded to an approximate dimension of 15-1/2 by 10 inches and inserted in a polyethylene bag. The bag shall be flat style, made of clear polyethylene film of 0.00125 inch thickness (± 20% tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.3 Level C. Trousers shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.



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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. One hundred pairs of trousers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a taped liner conforming to grade V3c of PPP-B-636. Level A packages shall be packed flat, two in length, one in width and fifty in depth within a shipping container. Inside dimensions of each container shall approximate 20 inches in length, 15-1/2 inches in width and 19 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Each containers shall be closed, waterproofed, and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level B. One hundred pairs of trousers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF or SF, variety SW, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a taped liner conforming to type CF, variety SW, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed flat, two in length, one in width and fifty in depth within a shipping container. Inside dimensions of each container shall approximate 20 inches in length, 15-1/2 inches in width and 19 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Each container shall be closed in accordance with method II, as specified, in the appendix of the container specification. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.3 Level C. Trousers packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.



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5.3.2 Polyethylene bagged packages. Polyethylene bagged packages shall have the required information legibly printed or stamped in a contrasting color directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag information shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
QUANTITY
SIZE

6. NOTES

6.1 Intended use. The green, cotton, man's, surgical, operating trousers covered by this specification are intended for wear by the male surgical personnel of the Medical Services. The trousers are used in conjunction with the shirts, operating, surgical, man's, cotton, sleeveless, style A conforming to MIL-S-37129 and shirts, operating, surgical, man's, cotton, quarter length sleeves, style B, conforming to MIL-S-37130.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Size and quantity desired (see 1.2).

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Supersession data. The green, cotton, man's, surgical, operating trousers covered by this specification supersedes the Item No. 2 trousers covered by MIL-S-36175A and Specification Data Sheet No. 16 for 6532-00-299-9628, 9629, 9630, and 9631.

6.4 Patterns. The patterns for the trousers will be furnished by the contracting officer to the contractor for use in cutting working patterns (see 3.3.1). The pattern parts shall consist of the following:

| NOMENCLATURE | CUT PARTS |
|---|---|
| Front | Two |
| Back | Two |



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6.5 Catalog Information. This specification is intended to cover the following items appearing in the Federal Supply Catalog:

| NATIONAL STOCK NUMBER | ITEM IDENTIFICATION |
|---|---|
| 6532-00-299-9628 | TROUSERS, OPERATING, SURGICAL, Men's, Size Large, Cotton, Green, Each |
| 6532-00-299-9629 | TROUSERS, OPERATING, SURGICAL, Men's, Size Extra Large, Cotton, Green, Each |
| 6532-00-299-9630 | TROUSERS, OPERATING, SURGICAL, Men's, Size Medium, Cotton, Green, Each |
| 6532-00-299-9631 | TROUSERS, OPERATING, SURGICAL, Men's, Size Small, Cotton, Green, Each |

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6532-1091
is



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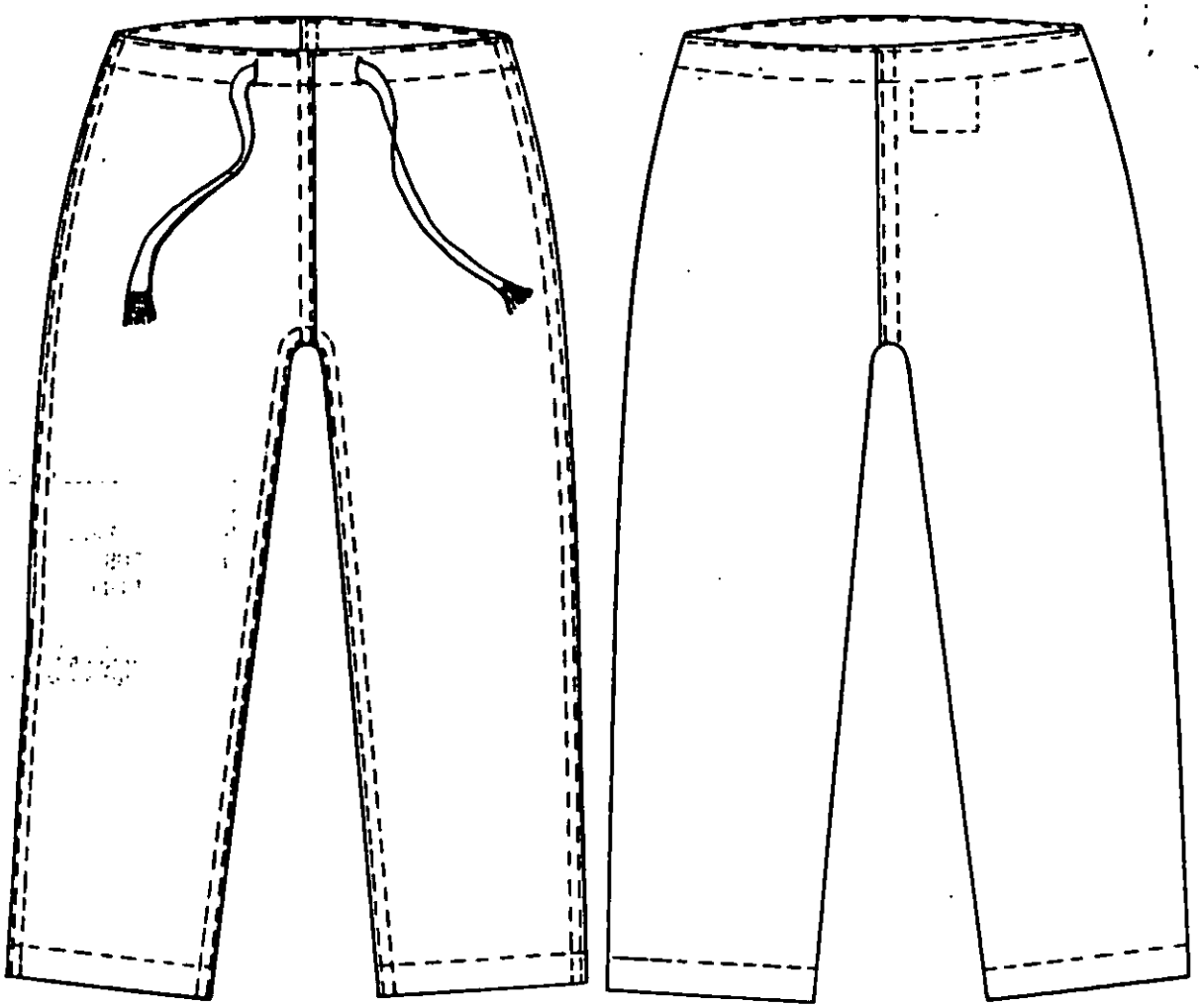



FIGURE 1



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-37131

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO. |
|---|---|
| | DATE |

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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